

## 4 Feature selection

### 4.1 Introduction

In machine learning, a proper selection of features can speed up a large variety of algorithms, oftentimes even increasing their performance due to the removal of noisy features. It is a combinatorial problem without an optimal solution [10]. Despite existing multiple kinds of algorithms performing this task, there is not a perfect solution for the combination of features that gives the perfect trade-off between computational burden and performance.

Using quantum systems for simulating and solving combinatorial problems is one of the main applications in current research in quantum computing and quantum optimization. In fact, mapping combinatorial problems onto quantum systems already has a number of real applications in industry. [8, 5, 11, 12, 2].

In this report, we propose a novel feature selection method based on quantum simulation of interacting Rydberg atom arrays. The intrinsic quantum behavior of this system together with appropriate encoding of importance and redundancy properties into arrangement and driving programming could lead to high-quality results outperforming classical methods.

### 4.2 Feature selection in Machine Learning

Feature selection (FS) is a fundamental technique in machine learning and data mining. Its primary goal is to identify an optimal subset of relevant variables (features) from the original dataset, aiming to reduce dimensionality while maintaining or even improving the predictive performance of the model. Effective feature selection can lead to increased model accuracy, reduced overfitting, and lower computational costs.

There are several strategies for FS, generally categorized as *filter*, *wrapper*, and *embedded* methods. Among them, filter methods evaluate each feature independently of any learning algorithm, typically using statistical criteria to assess the relevance of each feature. A widely used measure in this context is the **mutual information** (MI), which quantifies the statistical dependency between two random variables. Given an input variable  $X$  (e.g., a feature) and a target variable  $Y$  (e.g., a class label), the mutual information is defined as:

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \left( \frac{p(x, y)}{p(x)p(y)} \right), \quad (10)$$

where  $p(x, y)$  is the joint probability distribution of the input  $X$  and the output  $Y$ , and  $p(x)$  and  $p(y)$  are their marginal distributions. A high mutual information value indicates that knowledge of  $X$  provides significant information about  $Y$ , making  $X$  a potentially useful feature. In practical scenarios, FS often involves not only selecting relevant features individually but also minimizing redundancy among them. This leads to combinatorial optimization problems, where the objective is to select a subset of features that jointly maximize relevance to the target while minimizing redundancy. Such problems can be naturally mapped onto a *Quadratic Unconstrained Binary Optimization* (QUBO) formulation, where each binary variable represents the inclusion or exclusion of a feature. This mapping enables the use of emerging quantum and quantum-inspired hardware platforms, such as neutral-atom quantum devices, for solving FS problems efficiently.

### 4.3 Quantum encoding of classical feature properties

FS involves identifying subsets of variables from high-dimensional data that optimize predictive accuracy while reducing redundancy. Leveraging neutral atoms in Rydberg states, we introduce a mapping scheme where the redundancy between variables directly translates into spatial arrangements of atoms. Specifically, variables that exhibit higher redundancy are represented by atoms positioned closer together, resulting in stronger interactions.

The quantum system's dynamics are governed by a Hamiltonian that encodes both the redundancy between variables and their individual importance. Pairwise redundancy is represented through spatially dependent Rydberg blockade interactions, which effectively prevent simultaneous excitation of nearby atoms. Meanwhile, the individual importance or relevance of each feature is incorporated as node weights within the Hamiltonian, realized by adjusting local detuning parameters of the driving fields applied to each atom.

As explained above, we aim to simulate the FS cost function defined above. This expression would also be optimized by Gurobi algorithm for benchmarking purposes.

$$Q(x, \alpha) = -\alpha \sum_i^N I_i x_i + (1 - \alpha) \sum_{i,j}^N R_{ij} x_i x_j \quad (11)$$

- $I_i = I(x_i, y)$  is the individual importance with the target of each variable  $x_i$  and it is directly related with the individual weights  $w_i$  in MWIS problem.
- $R_{ij} = R(x_i, x_j)$  is the redundancy between variables and works as the 'interaction' between atoms,  $V_{ij}$ .
- $\alpha$  is a modulation parameter that balances both terms.

The correct shape and evolution of drivings is key to effectively drive the system towards the solution of interest. We can independently program the time evolution of  $\Omega(t)$ ,  $\Delta_g(t)$ ,  $\Delta_l(t)$  and  $\phi(t)$  parameters in the simulation to make the system energetically favorable to certain state combinations, while  $V_{ij}$  is controlled through atom positioning and establishes a constraint to those state combinations. Here we define a driving protocol that ensures an adiabatic evolution towards the ground state of the FS Cost function.

Starting from  $\Omega(t)$ , we should define a smooth driving profile that starts and ends in  $\Omega = 0$ , this ensures the addition of a transverse field which enables Hilbert space exploration and hence creates superposition among all possible state when  $\Omega$  is in its maximum value. Regarding the detunings, adiabatic principle establishes a connection from all  $|g\rangle$  state to FS solution when we slowly vary  $\Delta$  from a negative value to a positive one by assuming that we remain in ground state during the evolution. Since we can independently tune global and local detunings, we propose to divide the total evolution in a "two-step" protocol. We must start with all atoms in  $|g\rangle$  state, hence we start with a strong negative value of global detuning  $\Delta_g \ll 0$  and slowly vary towards  $\Delta_g = 0$  at half evolution, similarly for  $\Delta_l(t)$ , we start from  $\Delta_l = 0$  and slowly vary towards to  $\Delta_g \gg 0$  to effectively reproduce the cost function expression define above, as shown in Figure [12](#).

This gives the following expression for  $t = T$ , which is clearly analogous to eq. [11](#) by connecting local detuning term to individual importance and interaction term with redundancy between variables:

$$H_{Ryd}(T) = -\Delta_l^{max} \sum_i p_i \hat{n}_i + \sum_{i < j} V_{ij} \hat{n}_i \hat{n}_j \quad (12)$$

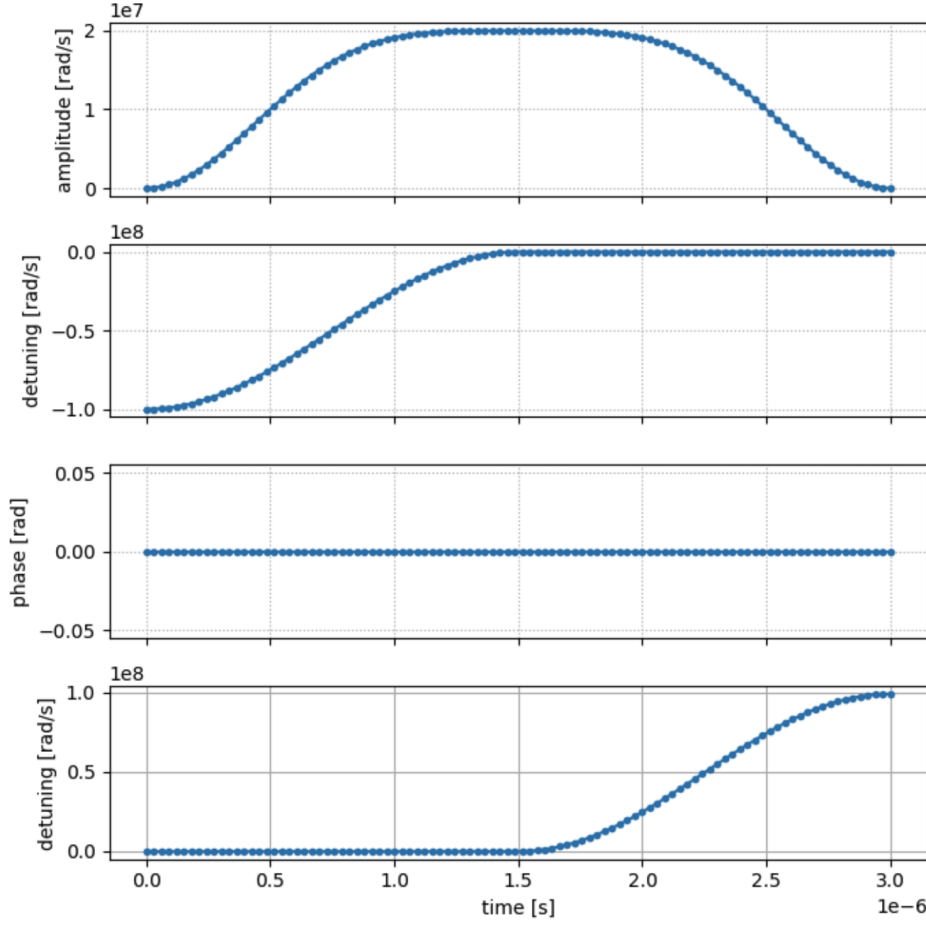


Figure 12: Function profiles for time-dependent parameters of  $H_{Ryd}$ , a smooth variation for all parameters is proposed, given by a two-step evolution for detuning, where half time of the evolution is governed exclusively by one of them.

#### 4.4 Atom Arrangement Based on Redundancy Matrix and Grid Mapping

Once the Hamiltonian has been connected to the cost function, the next step involves determining the spatial arrangement of atoms according to the redundancy matrix. Initially, we establish a mathematical relation between the redundancy values and the interatomic distances, defined as:

$$d_{ij} = \left( \frac{C_6}{R_{ij}} \right)^{1/6} \quad (13)$$

This formulation translates the level of redundancy between two variables into their spatial proximity: highly redundant variables are positioned closer to each other, while less correlated variables are placed further apart. However, this approach alone can lead to suboptimal arrangements when important but mutually redundant variables are forced into close proximity, potentially conflicting with the blockade mechanism and hindering an optimal selection.

To mitigate this issue, we introduce a second term in the distance calculation that accounts for the independent importance of each variable, using the parameter  $\beta$  to modulate its effect:

$$d_{ij} = \left( \frac{C_6}{R_{ij}} \right)^{1/6} + \beta(1 + I_i)(1 + I_j) \quad (14)$$

This adjustment increases the separation between atoms when both variables exhibit high independent importance, thereby reducing the risk of overlapping blockade regions for critical features. The parameter  $\beta$  is varied between 0 and 1 to balance redundancy and independent importance in the arrangement.

### Two-Step Arrangement Process

The atom arrangement is generated through a two-step procedure designed to ensure compatibility with hardware constraints and maximize the robustness of the layout:

#### 1. Distance Matrix Calculation:

- The modified distance matrix  $d_{ij}$  is computed using eq. 14 incorporating both redundancy and importance effects.
- This matrix is then used as input for a *Multi-Dimensional Scaling (MDS)* algorithm, which translates the distances into spatial coordinates while minimizing the overall error.

#### 2. Grid Mapping and Adjustment:

- After obtaining the initial MDS positions, the atoms are then positioned on a regular grid to adhere to hardware constraints.
- The grid spacing is automatically set to the minimum distance obtained in the first step, ensuring that the resulting layout maintains the relative proximities dictated by the redundancy matrix while respecting the hardware-imposed minimum spacing.
- This step effectively mitigates the risk of having interatomic distances below the Rydberg blockade radius, which could otherwise lead to unwanted double excitations.

Once the arrangement is prepared, we perform the driving protocol presented above for equidistant values of  $\beta$  parameter between 0 and 1 to observe how the individual importance contributes to the selection. The actual ranking of features is directly inferred from the measurement of Rydberg densities, which shows the population between state  $|0\rangle$  and  $|1\rangle$ . Once the Rydberg densities are computed, the choice of values is up to the user by choosing the most highlighted variables in the density histogram.

## 4.5 Results

### 4.5.1 Datasets and Experimental Setup

To evaluate the performance of the proposed algorithm, we tested it on two distinct datasets with different application contexts:

- **Telco Customer Churn Dataset:** Churn customer refers to the number of existing customers who may leave the service provider over a given period; these customers are called as churners. The objective is to identify the most relevant features that influence the customers to churn.

- **Oral Cancer Dataset:** This dataset consists of medical and clinical variables that may be associated with the diagnosis of oral cancer. The goal here is to identify a subset of measurable features that are most predictive of cancer presence, aiding in early detection and diagnosis.

For each dataset, the redundancy matrix was computed to establish the initial arrangement of atoms based on the inter-variable relationships defined in eq. 14. Figure 13 and Figure 14 illustrate the resulting atom arrangements for the Churn and Oral Cancer datasets, respectively. The arrangements were obtained using the two-step procedure described earlier, first generating the distance matrix and subsequently aligning the atoms to a hardware-compatible grid configuration.

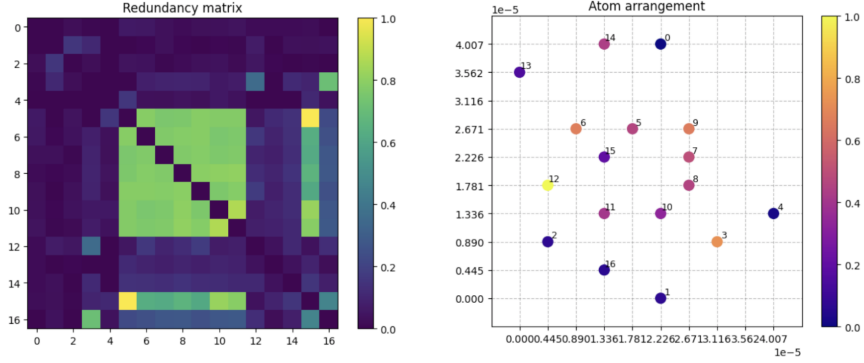


Figure 13: Atom arrangement for the Churn dataset, based on the redundancy matrix and grid mapping.

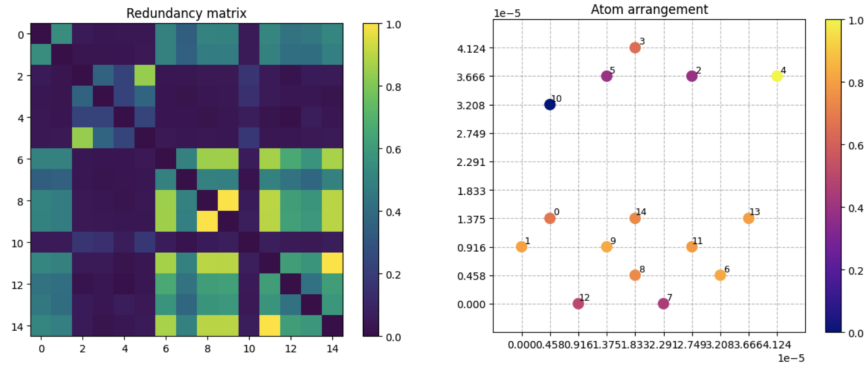


Figure 14: Atom arrangement for the Oral Cancer dataset, based on the redundancy matrix and grid mapping.

#### 4.5.2 Rydberg Densities for Varying $\beta$

Once the atom arrangements were established, the driving protocol was applied iteratively for different values of the parameter  $\beta$ , ranging from 0 to 1 in equal intervals. This allowed us to observe how the independent importance of features influenced the resulting Rydberg densities and, consequently, the selection of key variables.

Figure 15 and Figure 16 show the Rydberg densities obtained for each dataset across varying  $\beta$  values. By adjusting  $\beta$ , we can control the relative weighting of redundancy versus independent importance, allowing for more flexible feature selection strategies.

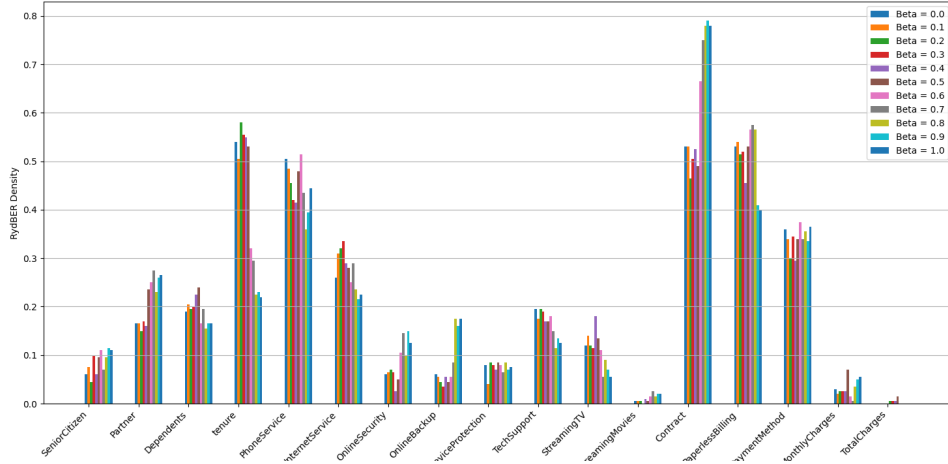


Figure 15: Rydberg densities for the Churn dataset across varying  $\beta$  values, showing the population of each state and its correspondence to selected features.

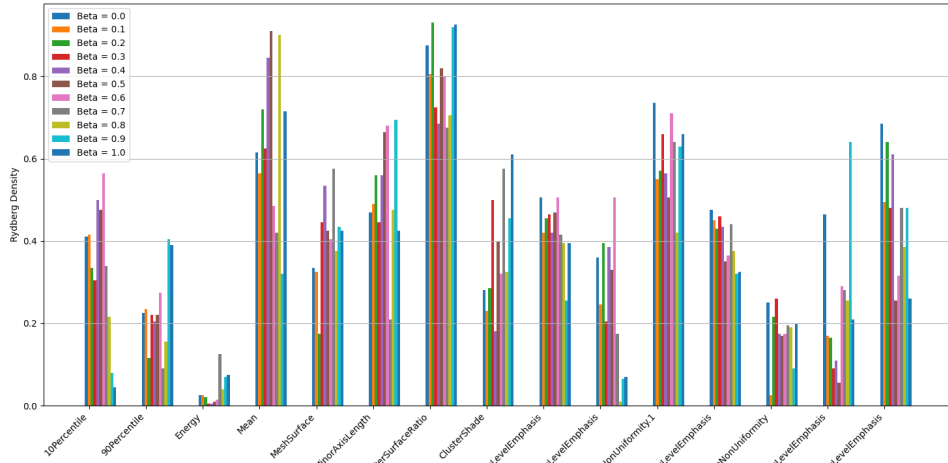


Figure 16: Rydberg densities for the Oral Cancer dataset across varying  $\beta$  values.

#### 4.5.3 AUC Analysis

The AUC results obtained for the Churn dataset are presented in Figure 17. While the overall AUC scores do not surpass those achieved with Boruta, it is noteworthy that similar performance levels are obtained with a significantly lower number of variables. Additionally, in several instances, our method outperforms the ranking-based selection, indicating the potential of  $\beta$  as a tuning parameter to balance redundancy and feature importance.

A particularly interesting observation is the plateau in AUC scores observed beyond 7 variables. This suggests that Boruta may have overestimated the number of relevant features, incorporating variables that do not contribute to a substantial increase in predictive power. Thus, our method provides a more efficient selection strategy, identifying a more compact and effective subset of variables.

The AUC results for the Oral Cancer dataset, presented in Figure 18, demonstrate even more promising outcomes. In this case, our method not only outperforms the ranking-based approach for most values of  $n_{var}$ , but also surpasses the AUC achieved by Boruta while using a smaller subset of

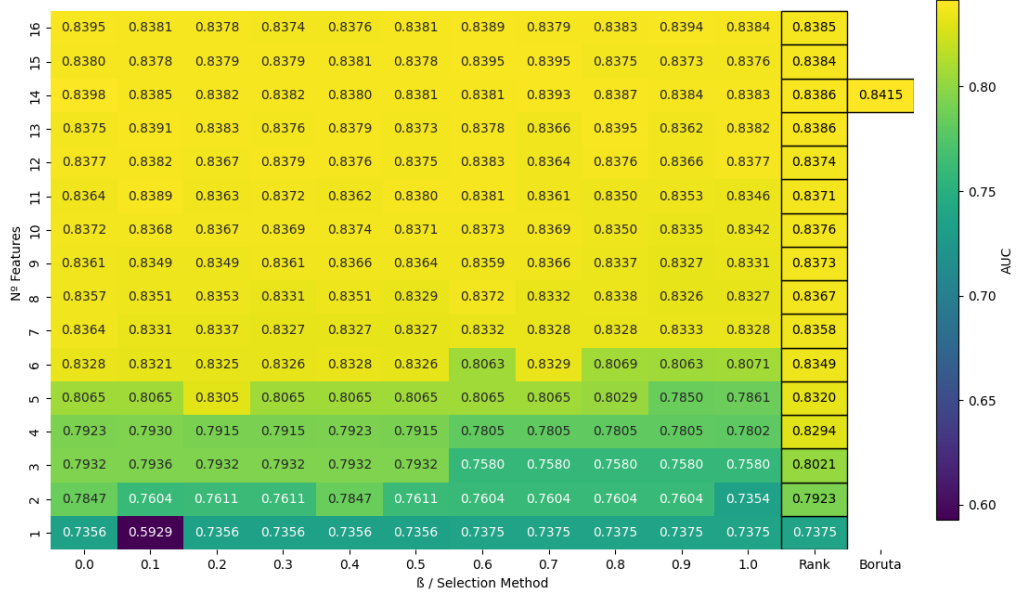


Figure 17: AUC comparison across different  $\beta$  values for the Churn dataset, alongside Rank-based selection and Boruta.



Figure 18: AUC comparison across different  $\beta$  values for the Oral Cancer dataset, alongside Rank-based selection and Boruta.

variables. This highlights the ability of our method to effectively capture relevant features without introducing unnecessary redundancy.

Moreover, the influence of  $\beta$  is more pronounced in this dataset, as certain values of  $\beta$  consistently lead to higher AUC scores across a wide range of variable counts. This emphasizes the importance of intelligently selecting the  $\beta$  parameter to maximize predictive performance.

## 4.6 Conclusions on Feature Selection

The parameter  $\beta$  plays a crucial role in modulating the contribution of each pair of variables to the overall distance matrix. By adjusting  $\beta$ , we effectively control the balance between redundancy and feature importance, as it introduces a secondary distance component that scales with the mutual information between each pair of variables and the target variable.

One promising approach currently under investigation is treating  $\beta$  as a hyperparameter during the training phase. By selecting the optimal  $\beta$  value for each subset size, we aim to maximize AUC scores without introducing unnecessary features. Future work will focus on optimizing  $\beta$  as a hyperparameter, adapting it to each dataset’s characteristics.

As a future work, we also plan to include more datasets with different characteristics in the analysis, in order to evaluate the robustness and applicability of our approach properly. The method should also be run in real neutral-atom hardware. We also plan to extend the idea to HUBO (Higher-order unconstrained binary optimization) problems, tailoring it to another hardware technology, namely, superconductors.

The proposed method introduces a novel approach to feature selection through Rydberg-based quantum encoding, integrating a flexible parameter  $\beta$  to modulate feature interactions based on mutual information. Summing up:

- In the Churn dataset, our method achieves comparable AUC scores to Boruta with a significantly smaller subset of variables, indicating more efficient feature selection.
- For the Oral Cancer dataset, the proposed method not only reduces the number of selected features but also improves AUC scores, demonstrating its effectiveness in handling highly redundant datasets.
- The parameter  $\beta$  emerges as a key factor in fine-tuning the balance between redundancy and independent importance, providing a powerful mechanism to adapt the feature selection process to the dataset’s underlying structure.

Overall, our method provides a promising alternative to classical feature selection approaches, leveraging quantum-inspired encoding to optimize the selection process while maintaining competitive predictive performance.